

STAGE 4 / TECHNOLOGY

Swinging Sign

STUDENT WORKBOOK



Name _____ Class _____

50 pages • A4 • 25 sheets duplex • Edition 1.0 • Course companion

Illustrative cover. Use the teacher-issued plan for construction.

THE LEARNING PATHWAY

Your learning pathway

Use the course for demonstrations and media, and this book to think, practise and record evidence.

Part	Learning focus	Pages
Start here	Course, pathways and practical information	1–4
Module 1	Brief and readiness	5–12
Module 2	Communicate the design	13–20
Module 3	Plan for production	21–28
Module 4	Produce and control quality	29–36
Module 5	Finish, test and evaluate	37–44
Project folio	Brief, concepts, proposal, process and evaluation	45–50

Read and apply

Each module explains the ideas before asking you to use them. Look closely at photographs and diagrams, follow the worked reasoning, then try the paper tasks. The questions in this workbook are formative adaptations of the course learning. Your teacher may also use the website's larger knowledge checks and scaffolded responses.

Build the evidence as you work

Use the module activities as practice and keep your own project decisions in the folio. Cross-reference a useful module page instead of copying a whole answer. Record changes and the reasons behind them while they are fresh. A final photograph cannot show every decision that led to a product.

Matching course <https://stevenecowell.github.io/Swinging-Sign/>

Student edition 1.0 • 50 A4 pages • 25 sheets when printed double-sided on the long edge. Use the PDF for controlled printing. The Word edition keeps text and working fields editable; editing can change pagination.

START HERE

Know what guides your work

The workbook develops design thinking alongside your teacher's current instructions.

This companion follows the five modules of the Stage 4 Technology Swinging Sign course, in Materials and production processes. The Swinging Sign is the central project. The LED tea-light holder and scrolled pot plant holder are alternate or additional pathways only when directed by the teacher. Use the pathway that has actually been approved for your class.

Learning and practical authority

The reading explains relationships, choices and evidence. It cannot authorise practical work. Before acting, use the teacher-issued plan, current demonstration, local risk controls and specific permission. The reference photographs show appearance; they do not confirm stock, dimensions, joints, settings, loads, mounting or an approved making method.

A drawing in Onshape communicates a proposal for review. A complete-looking model or file is not proof that its dimensions, process or support system have been approved. Keep the current version identifiable and record what the teacher has confirmed before proceeding.

Evidence you can trust

Record what you observed, the conditions, the relevant criterion and the decision that followed. Label an example as an example and an unknown as a question. Write "not yet checked" when evidence is missing. Never fill a process log with imagined practical work or a test result that was not obtained.

Use the formats together

Readings and paper activities can be completed with the teacher presenting the matching course. The website provides no-stream learning routes as well as optional media. Your teacher controls any required CAD session or practical demonstration. The book's responses support learning; the current assessment notification supplies any formal conditions, criteria, due dates and marks.

At a hold point Pause if the plan, task, controls, version or permission is unclear. Record the question and ask the teacher; uncertainty is useful information to resolve.

Identify one practical detail you need confirmed and explain why the photograph or model cannot confirm it.

START HERE

Compare the project pathways

A different product can use the same careful design process.



Figure 1. Original illustrative comparison: sign, LED tea-light holder and scrolled pot plant holder. Compare visible purpose and form; these examples do not specify what your class will make.

Pathway	Purpose to investigate
Swinging Sign	Communicate an agreed message to an intended audience.
LED tea-light holder	Hold and present a battery LED light source.
Scrolled pot-plant holder	Support or contain a pot within teacher-confirmed conditions.

All three pathways need a clear brief, explained ideas, responsible material choices, planned evidence and evaluation. Their specialised requirements can differ. Do not assume that one is easier, that all three must be made, or that a similar appearance permits the same process.

Record your teacher-confirmed pathway, intended user and one question about its conditions of use.

READ / OBSERVE / APPLY

Read the need before the shape

A useful design begins with a person, a purpose and a place.

A sign helps an audience recognise a place or message. The owner is one user, but the people who need to read it are users too. Begin by naming the job it must do and the people it must help. A decorative object without a clear purpose gives you little basis for choosing between ideas.

Context explains the situation around the product. Think about where viewers approach it, how long they have to look, the background and the conditions of use. A quiet indoor display and a sign noticed briefly from farther away may need different priorities. These questions guide design thinking; they do not supply dimensions or a mounting method.

Form is the visible and physical character of the product: shape, line, proportion, colour, texture and arrangement. Function is what the product does. Repeated curves can connect the bracket and panel visually, while a clear message can help the audience recognise the purpose. A pleasing form still needs to work for its intended audience.

Analyse a product by linking a visible feature to a likely effect. Write what you can observe, then explain how it might help the user. A photograph may show that a panel hangs below a bracket; it cannot establish the exact stock, hidden connection or safe load. Record those unknowns and use an authorised source before making a practical decision.



Figure 2. Illustrative course reference. The complete bracket and blank oval panel provide visible evidence of form; the image is not a construction plan.

Name one audience and explain one feature that could help this audience recognise the intended message.

INVESTIGATE / APPLY

Observation or assumption

Describe the evidence before explaining what it might mean.

An observation reports what is visible. An inference is an explanation that goes beyond direct observation. A question needing confirmation identifies missing information. The categories help you think clearly, but none gives permission to act on an unsupported practical claim.

Worked distinction “The panel is oval” describes visible form. “The hidden joint was made by a particular process” goes beyond what the image can show. “What load is approved?” identifies information still needed.

Use the course reference on page 5 and the illustrative pathway comparison on page 4. Classify each statement below as observation, inference or needs confirmation. This paper task adapts the course evidence-sorting puzzle.

Statement	Category
The oval panel hangs below a bracket.	
The hidden joint was made by one exact process.	
The approved load for the pot holder.	
The tea-light holder has a geometric hanging form.	
The project uses a particular stock thickness.	

Explain the categories you chose. Why must an inference about construction be checked before it becomes a specification?

Write one product-analysis sentence linking a visible feature to a likely effect on a user.

READ / OBSERVE / APPLY

Give the brief useful boundaries

Criteria explain success; constraints define the confirmed limits.

A design brief connects a situation, an intended user, a need and a proposed response. “Make something from metal” gives little direction because almost any object could fit. “Design a sign that helps visitors recognise an agreed place” gives the designer a purpose while allowing different ideas to be explored.

Criteria are the qualities you use to judge success. They may concern message recognition, the character of the design, clear communication of the proposal or responsible resource use. A useful criterion identifies something that evidence can address. “Looks good” is difficult to test because it hides whose view matters and which qualities are being judged.

Constraints are confirmed boundaries such as available time, resources, approved processes and teacher-issued project requirements. A missing stock size is not a constraint you can invent; it is a question to resolve. Keep the brief focused on the purpose while recording the practical details that still need teacher confirmation.

Criteria can compete. More decorative detail might strengthen a theme while making a message harder to recognise. A trade-off is a reasoned choice between effects, not simply an error. Rank the qualities that matter most for this audience and context, then explain what you are willing to compromise. The first attractive idea is a starting point for comparison, not the automatic final choice.

A brief that leaves room to design

Try this structure: design a teacher approved product for an identified audience that addresses a stated need in a stated context. Separate what the project must achieve from unconfirmed information about how it will be made.

Draft a brief for your approved pathway and name two criteria that could compete. Explain the trade-off.

WORKED EXAMPLE

Turn a preference into a criterion

A criterion needs a quality to judge and evidence to examine.

Worked example

Imagine a class designing a sign for a quiet reading corner. The first proposal says, “It should look amazing.” This expresses a preference but gives a reviewer no clear way to judge success. Different people may mean different things by “amazing”.

A stronger criterion The intended audience can identify the agreed message during a teacher approved viewing check.

This version identifies the audience, the communication quality and a possible source of evidence. It leaves the viewing method and conditions for teacher approval. A second criterion could ask whether the visual character suits the reading corner. These criteria can compete if decoration makes the message harder to recognise.

Apply the distinction

For this fictional case, the teacher has confirmed a deadline and permitted processes. Sort the statements as criterion, confirmed constraint or question needing confirmation. Write the category in the second column.

Statement	Category
The intended audience can identify the message.	
Only the teacher-approved processes may be used.	
What stock sizes will the teacher supply?	
The proposal must be ready by the confirmed deadline.	

Explain your sorting and rewrite one vague criterion for your own intended audience.

Reasoning move State what matters to the audience, then explain what observation could help you judge it. A criterion is not permission to run a practical test.

READ / OBSERVE / APPLY

Be ready before you act

Readiness combines understanding, current conditions and permission.

A hazard has the potential to cause harm. Risk describes the chance and possible seriousness of harm in a particular situation. A control is an action or condition used to reduce risk. The same hazard can create different risk when the task, people, surroundings or controls change. Knowing the word “hazard” is only the beginning of responsible decision-making.

A readiness check asks whether you understand the goal, have the approved plan, know the current instruction and controls, and have permission to begin. It also asks whether the work area matches what the teacher expects and what evidence should be recorded. If a condition or instruction is unclear, pause and ask before acting.

Personal readiness matters too. You need to concentrate, communicate and respond to instructions. Rushing to catch up or copying another group can hide a misunderstanding. Yesterday’s setup or an online example may not match today’s task. A classmate’s confidence is not evidence that you have the same authorisation.

Protective equipment is one part of the local control system. It does not replace a demonstration, an approved process, supervision or a check of the current situation. Record the question you need answered rather than improvising a procedure. Calm preparation also protects materials and time by helping prevent avoidable mistakes.



Figure 3. Illustrative workshop still life. Protective equipment and a notebook prompt preparation; the photograph is not a complete list of controls or an operating procedure.

A student says, “I did something similar last week, so I can start now.” Explain the missing readiness checks.

INVESTIGATE / APPLY

Trace a responsible choice

Sustainability decisions begin before production and continue through use.

Life-cycle thinking follows a product through sourcing, production, use, maintenance and end of life. Recyclability matters, but it is one consideration among material quantity, waste, durability and effects on people and environments. A design can waste resources long before someone decides how to dispose of it.

Explain a responsible choice as a chain: identify a concern, compare options, choose for a reason and state what evidence could show the effect. Reducing unnecessary decorative features might reduce material use, but only if the intended function and agreed criteria are still met. Keep the expected effect separate from a result you have actually observed.

For each stage, record one useful question about your proposed product. Do not invent a local material, process or disposal rule.

Life-cycle stage	Question to investigate
Sourcing	
Production	
Use	
Maintenance	
End of life	

Choose one life-cycle concern. Explain an option, a trade-off and evidence to seek. Connect it to one readiness point on page 9: how could the two change your proposed decision?

Terms in context Durability concerns lasting performance in use. Maintainability concerns how a product can be cared for or repaired. Both affect its useful life.

KNOWLEDGE CHECK

Check the design reasoning

Circle one best answer for each question. Use the module readings to explain your choices.

1. Which statement uses visible evidence responsibly?

- ☐ A. The bracket is safe for every possible load.
- ☐ B. The panel hangs below a bracket; its approved load still needs confirmation.
- ☐ C. The photograph tells us the exact stock size.
- ☐ D. Every hidden connection was made by the same process.

2. Which statement is a criterion rather than a confirmed constraint?

- ☐ A. Only approved processes may be used.
- ☐ B. The teacher has confirmed the available time.
- ☐ C. The issued plan must be followed.
- ☐ D. The intended audience can recognise the message in an approved check.

3. What should happen when an instruction or permission is unclear?

- ☐ A. Pause and ask the teacher before acting.
- ☐ B. Copy another student who appears confident.
- ☐ C. Choose an online procedure and begin.
- ☐ D. Assume yesterday's conditions still apply.

4. Which sustainability explanation considers the whole life cycle?

- ☐ A. The product is metal, so it has no environmental effects.
- ☐ B. Its colour alone proves that it is sustainable.
- ☐ C. Its material use, production, useful life, maintenance and end of life all need evidence.
- ☐ D. Recyclability means durability does not matter.

Choose one incorrect answer and explain the assumption behind it.

PROJECT FOLIO

Compare products with a purpose

Use a common question to make the comparison fair.



Figure 4. Illustrative reference B. Compare this panel and bracket with reference A on page 5. Neither image supplies approved construction information.

A comparison is more useful when it examines the same factor in both products. Consider how the outline and arrangement could affect an audience, then identify what each image leaves unknown. A different form is not automatically better; its value depends on the intended purpose and context.

Record one observation for each reference, explain a likely effect, and identify a question needing an authorised source.

Reference	Observation and likely effect	Question to confirm
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A • page 5

B • this page

What useful design idea could you investigate from this comparison without copying either product as a plan?

LOOK CLOSELY

Two ways to build an image

Choose the structure that serves the message.

A digital photograph and a clean sign symbol may look similar on a small screen, yet store their information differently. A bitmap, also called a raster image, uses a grid of coloured pixels. Each pixel contributes a small part of the image. This suits photographs because many subtle changes in colour, texture and tone can be represented together.

A vector graphic describes objects using points, lines, curves and shapes. When a vector is enlarged, the software recalculates these paths. The boundary can remain smooth at different sizes, and individual objects remain available for editing. This makes vectors useful for clean lettering, logos and silhouettes. Enlarging a bitmap spreads the available image information across a larger display; it does not recover detail that was never stored.

Choose according to purpose. A presentation about a proposed sign might use a bitmap photograph to show its setting and vector objects to develop the message. Neither format is automatically superior. The question is whether you need photographic detail, editable shapes or both. A file extension is only a clue: a bitmap placed inside a vector document can remain a bitmap.

Automatic tracing attempts to turn visible bitmap boundaries into paths. It may also preserve unwanted backgrounds, uneven contours, overlaps, tiny gaps or unnecessary points. Treat the result as a draft. Compare its positive shapes and negative spaces with the design intention, check readability and simplify only where the meaning is preserved. Use original or teacher approved source material and confirm permission to reuse it. A successful conversion still needs the teacher's file review; it does not make a production-ready design.

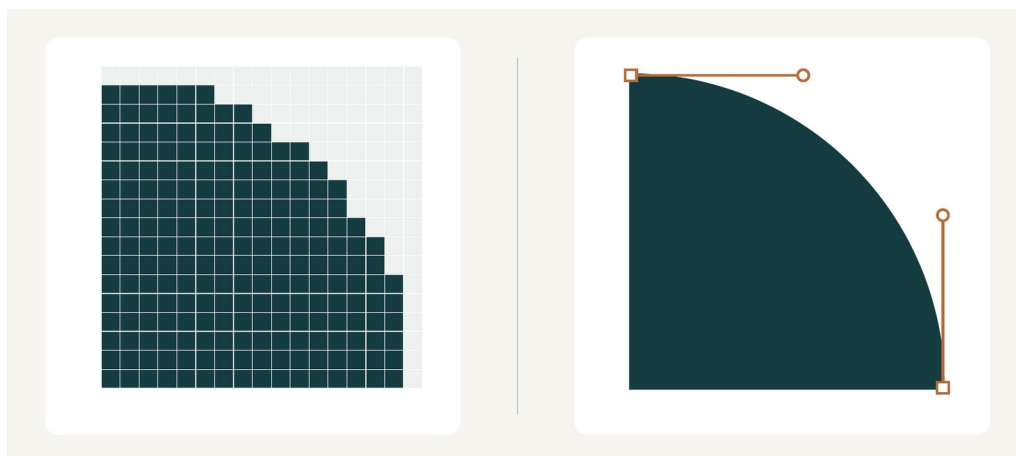


Figure 5. Concept comparison: the bitmap uses a pixel grid; the vector uses a path controlled by points. Enlargement changes their visible edges differently. This is a learning diagram, not a production file.

READ / OBSERVE / APPLY

Give an Onshape sketch a purpose

Control the relationships before developing the form.

Onshape is the CAD program used in this course. CAD means computer-aided design: the model records a design so it can be inspected, developed and revised. In an Onshape Part Studio, a sketch begins on a selected plane or planar face. That choice establishes the sketch's orientation and its relationship to later features. Begin by asking which view best explains the intended form.

Sketch geometry includes entities such as lines, circles and arcs. A dimension controls size or position. A geometric constraint controls a relationship: for example, whether two points meet, two lengths are equal or a line stays horizontal. Together, these controls express design intent. The aim is to make the sketch behave for a reason, rather than merely look correct at one moment.

Read the sketch feedback and investigate what it means. In the course's Onshape guidance, blue geometry indicates that control is incomplete, black indicates fully constrained geometry and red indicates a constraint problem. Colour alone is not enough. Check what can still move and inspect the displayed relationships. When a conflict occurs, look for competing or duplicated controls instead of deleting them at random.

A feature uses sketch geometry to create or change a three-dimensional part. The feature list records this sequence, so an earlier edit may affect later geometry. Save meaningful versions and write a short note explaining important changes. This helps a reviewer understand both the current model and the decisions behind it. A convincing model still requires teacher review. The teacher confirms document location, units, approved dimensions, sharing, drawing and export requirements, and any permission to move beyond design work.

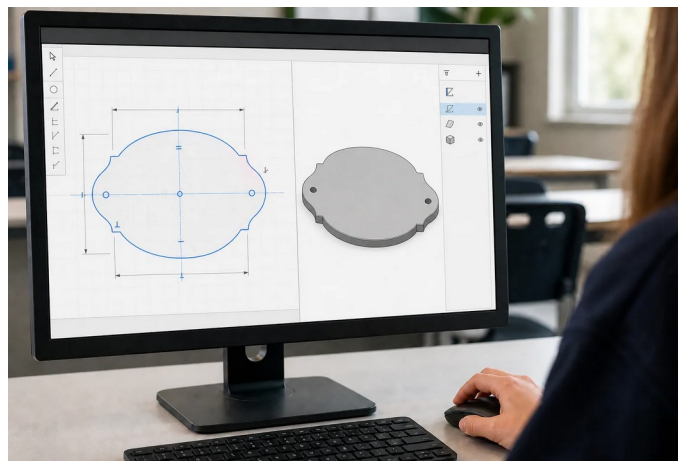


Figure 6. Conceptual CAD illustration: a sketch and a proposed form are shown together for review. This is not an Onshape interface guide or a construction specification.

READ / OBSERVE / APPLY

Make the message easy to find

A viewer should understand more than the designer's intention.

Legibility is how easily a viewer can recognise letters, symbols and important forms. For a sign, the intended message needs to reach its audience. Letter style, spacing, contrast and surrounding detail work together. A beautiful motif can weaken the communication if it competes with the main word. Negative space — the areas around and within forms — helps separate elements and makes their relationships clearer.

Visual hierarchy gives the viewer an order: what should attract attention first, then next? Size can establish emphasis, but so can contrast, position and spacing. Give the main message a clear role and make secondary details support it. Context matters. A sign glimpsed briefly may need a different arrangement from a page read closely. Use the teacher approved viewing check to investigate communication, rather than inventing a production or installation test.

Annotations make reasoning visible. A label names a feature; an annotation connects the feature to a reason and expected effect. "Open space around the main word should help it stand apart from the motif" is more useful than "space". It identifies something that a reviewer can examine. Mark unconfirmed material, mounting or production information as a question instead of presenting it as a settled decision.

Peer review gives another viewer a chance to test the intended effect. Notice a specific feature, connect it to the brief or a criterion, then suggest a focused investigation. The designer considers the evidence and may accept, modify or reject the suggestion with a reason. When comparing revisions, keep the viewing conditions consistent and change one major variable where possible. Otherwise, it becomes difficult to tell which change improved the result. Keep the earlier version and explain the decision so the development remains traceable.

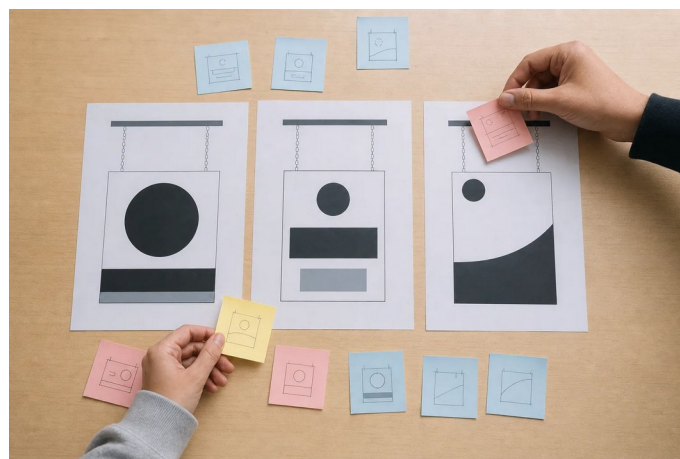


Figure 7. Illustrative design-review scene: compare hierarchy, spacing and the relationship between message and motif. This is communication evidence, not an approved design.

WORKED EXAMPLE

Control the movement you can explain

Paper Onshape clinic • fictional sketch state

A practice sketch lies on a teacher-supplied front plane. One line has its left endpoint fixed at the intended reference. Its free endpoint can move, so the line can rotate and change length. The design intention is for the line to remain horizontal with a teacher approved length.

The useful question is not “How many constraints can I add?” A horizontal relationship addresses rotation; the approved length dimension addresses length. Each has a different purpose. If the model reports a conflict, inspect existing controls before adding another. A fully constrained state is useful only when the controls express the intended design.

Explain the two controls in the example and the movement each is intended to remove.

A later feature uses this sketch. Explain why changing the sketch may affect that feature, and why the feature list matters.

Write a version note for this fictional revision and two questions for teacher review.

KNOWLEDGE CHECK

Check how the design communicates

Circle one answer for each question.

1. A sign presentation needs a detailed setting photograph and editable clean lettering. Which choice is best justified?

- ☐ A. Use a bitmap photograph and vector lettering.
 - ☐ B. Convert both to tiny screenshots.
 - ☐ C. Assume every file must contain only vectors.
 - ☐ D. Use whichever extension has the shortest name.
-

2. A new Onshape constraint creates problem feedback. What is the most useful first response?

- ☐ A. Delete every earlier control.
 - ☐ B. Inspect the new and existing relationships for conflict.
 - ☐ C. Ignore the feedback if the outline looks right.
 - ☐ D. Export the sketch to avoid reviewing it.
-

3. Which note is an annotation rather than only a feature label?

- ☐ A. Main word
 - ☐ B. Curved border
 - ☐ C. The open space separates the message from the motif so it is easier to recognise.
 - ☐ D. Dark shape
-

4. What makes a comparison between two design revisions more informative?

- ☐ A. Change every variable and the viewing conditions.
 - ☐ B. Ask only whether the designer likes it.
 - ☐ C. Discard the earlier version immediately.
 - ☐ D. Compare one purposeful change against the same criterion and viewing conditions.
-

DRAW / EXPLAIN

Explore a clearer reading order

Legibility lab • work entirely on paper

Use an original short message for a fictional school display. Create two layouts with the same words and motif, changing one major variable such as spacing or emphasis. These are communication concepts, not approved making plans.

Left: Concept A, with your intended first-second-third reading order. Right: Concept B, changing one major variable and adding a feature-reason-expected effect annotation.

Concept A	Concept B
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Use the same viewing approach for both. Record a strength, an issue and one notice-connect-suggest comment.

INVESTIGATE / APPLY

Look inside the format claim

Format triage • fictional evidence cards

Card	What is known
A	A photograph contains subtle surface textures and becomes blocky when enlarged.
B	Lettering consists of individually editable curves and points.
C	An image is inside a file called “vector design”; no object structure has been inspected.
D	A symbol can be enlarged smoothly and its outline points can be adjusted.

Classify A–D as likely bitmap, likely vector or needing more evidence. Give a clue and a suitable use for each.

If A were traced for a sign motif, predict two problems to check. Add one source-permission question.

Explain why one presentation could sensibly contain both vector and bitmap images.

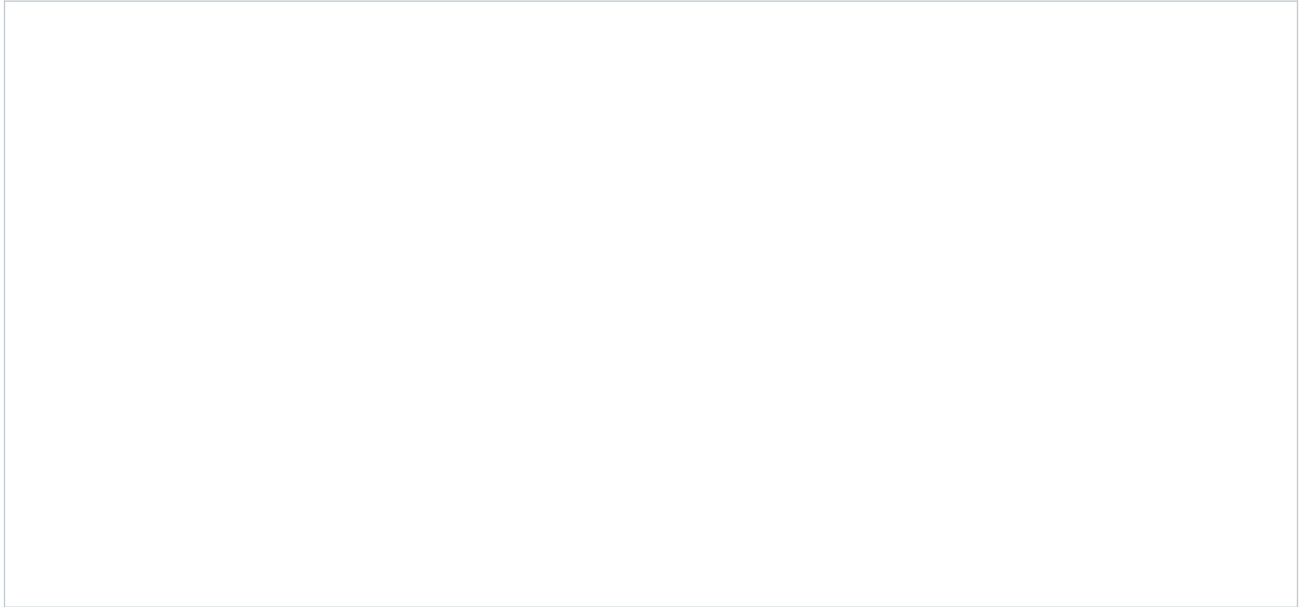
PROJECT FOLIO

Show the proposal and its reasoning

Folio links • concept options and final annotations

Record your selected communication proposal, using your own sketch or permitted evidence. State whether it is a current Onshape version, a paper concept or a planned revision. Include unresolved questions rather than inventing approval.

Show the proposal. Annotate hierarchy, one important design relationship and information still awaiting teacher confirmation.



What feedback did you consider? Explain one change you made, modified or rejected and why.

Identify the version or sequence label and explain what evidence makes it ready for review. Name one review question.

READ / OBSERVE / APPLY

Choose properties for a purpose

A material decision is a reasoned comparison.

A material property describes how a material behaves. Strength concerns resistance to forces that could pull, push or bend it. Hardness concerns resistance to scratching, denting or wear. Toughness concerns absorbing energy without breaking suddenly. These are different questions: resisting a surface mark does not automatically mean resisting a sudden break. Ductility and malleability describe different ways material can change shape without cracking. Corrosion resistance concerns how it withstands chemical attack from its environment.

The relevant properties depend on the product's purpose and context. A sign needs to communicate, keep its intended form and suit its approved display conditions. Mass, stiffness, surface appearance and environmental exposure may all matter. The course source names aluminium and steel sheet as possibilities for one sign pathway, but that does not identify the photographed material or approve stock for your project. The teacher confirms the options, their form and their practical suitability.

Compare a group of properties rather than searching for one “best” material. A lower-mass option may make handling easier, while another may better retain the intended shape. A trade-off is a deliberate compromise between competing benefits or limitations. Decide which criteria matter most, examine reliable evidence and acknowledge what is unknown. A score such as “good” says little unless the reason behind it is explained.

A useful justification links need, property, evidence and decision. It may recommend a material conditionally, while asking the teacher to confirm stock or the mounting system. Include environmental thinking: likely service life, maintenance, avoidable material use, useful offcuts and the school's end-of-life pathway. The alternate LED tea-light and pot plant tasks have their own requirements; material and support decisions cannot be copied from the sign. LED-only use and teacher-confirmed pot loading remain essential boundaries.

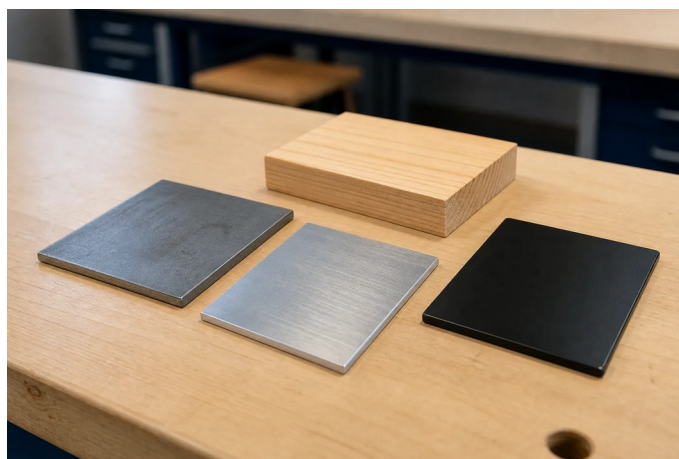


Figure 8. Illustrative material comparison: appearance invites investigation but does not establish material identity, grade, thickness or suitability.

READ / OBSERVE / APPLY

Make the layout checkable

Clear measurements and thoughtful arrangement reduce avoidable error.

A dimension communicates an intended size or position. Its unit gives the number meaning, while a datum identifies the reference from which the measurement is understood. These parts belong together. An unexplained number or a measurement taken from an unknown edge makes a proposal difficult to check. Keep the teacher approved values, units and references consistent across the Onshape drawing and planning notes.

Accuracy and precision are related but different. Accuracy concerns closeness to the intended value. Precision concerns detail or repeatability. Several readings can be very similar to one another yet all differ from the intended value. Repeating a measurement without checking the reference can therefore repeat the same mistake. Review the information as well as the number.

A layout shows how the required shapes occupy the available area. Check that all parts are present, their orientation is intentional and the arrangement stays within the confirmed stock boundary. Nesting explores how those shapes can use the area thoughtfully. It is not simply packing them as tightly as possible: approved spacing, edge requirements, orientation and process allowances still come first. Those requirements must be supplied locally.

A paper model helps separate the idea from workshop specifications. On a ten-by-ten grid, shapes covering sixty squares occupy sixty per cent of the area. The remaining forty squares are not automatically waste. A connected blank region may represent a useful offcut, whereas scattered narrow gaps may be less useful. Compare clarity, resource use and the risk of rework together. A layout that saves area but obscures the required parts may be a weak proposal. Record the chosen version and the reason for the change so another reader can review the plan.

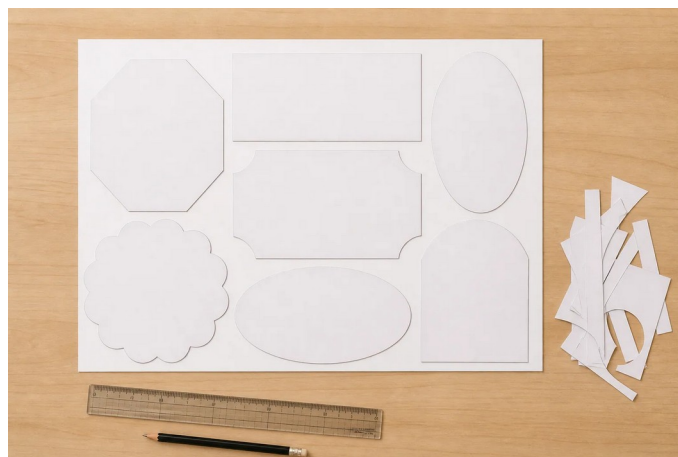


Figure 9. Paper-layout concept: compare required shapes, unused area and connected offcuts. The image supplies no project spacing, stock size or process allowance.

READ / OBSERVE / APPLY

Present one traceable proposal

A drawing, its version and its approval must agree.

An Onshape drawing communicates model geometry through selected views, dimensions and annotations. It allows the teacher to compare a proposal with the design criteria. The drawing should agree with the current model, but a polished page does not establish that every practical detail has been approved. The teacher supplies the required template, units, scale convention, dimensioning and export requirements.

Version control helps everyone identify which design is being discussed. Keep a meaningful project identifier, a version or date and a concise revision note using the teacher's naming system. Saving over all earlier work can erase the decision trail. Keeping many similarly named files can make the current state unclear. A review record should show which version was checked, what changed and which version is now awaiting or carrying approval.

Mounting belongs in the proposal because it affects orientation, balance, clearance, appearance and the path of forces to the support. A photograph may suggest a visible arrangement but cannot prove hidden connections, load limits or an installation method. Explain the intended context and relationship between product and support, then list what the teacher must confirm. The LED tea-light alternative remains LED-only; the pot plant alternative requires its own confirmed pot, loading and stability information.

Teacher approval is a hold point: a planned pause before an action becomes difficult to reverse. Bring together the current design view, version record, criteria check, material reasoning, layout, mounting concept and unresolved questions. If feedback requires a change, record it and submit the revised version for review. Silence does not mean approval, and approval of an earlier version cannot be assumed to cover a later change. The teacher communicates the authorised next stage; the student's job is to make the decision easy to trace.

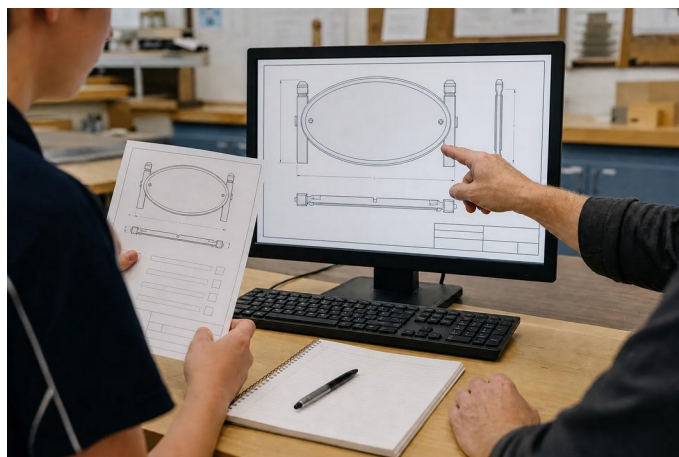


Figure 10. Illustrative review pack: drawing, version, criteria and questions need a clear connection. No approval is granted by this image.

WORKED EXAMPLE

Audit the pack before the hold point

Fictional proposal • identify present, unclear and missing evidence

Pack item	Available record
Drawing	Preview labelled revision B
Version note	Describes a spacing change from A to B
Criteria check	Completed against revision A
Layout	Revision B is identified, but required spacing is not confirmed
Mounting	“We will decide after making”; no reviewed proposal

This pack contains useful evidence, but its parts do not yet tell one consistent story. The drawing and revision note identify B. The criteria check still refers to A, so it does not show whether the change meets the criteria. The layout identifies a version while leaving an important local requirement open. The mounting relationship is missing from the design review.

A reasonable next action is to revise the pack and submit it for review. Opening the drawing is file access; assembling clear evidence is review readiness; receiving explicit teacher permission is production authorisation. These are different states.

Write three precise review questions that address the unresolved evidence.

Explain why this proposal should be revised and reviewed before production. Name one item that is already useful.

KNOWLEDGE CHECK

Check the planning argument

Circle one answer for each question.

1. What makes a material justification stronger than “this metal looks best”?

- ☐ A. Choosing the most familiar name
 - ☐ B. Giving every option the same unexplained score
 - ☐ C. Linking a confirmed need to properties, evidence and a trade-off
 - ☐ D. Copying the material from an unverified photograph
-

2. Repeated measurements are nearly identical but all miss the intended value. What does this illustrate?

- ☐ A. Precision guarantees accuracy.
 - ☐ B. The unit no longer matters.
 - ☐ C. A datum is only decoration.
 - ☐ D. Repeatability does not by itself establish accuracy.
-

3. Two paper layouts cover the same area. Why might one still be more useful?

- ☐ A. It leaves a connected offcut while meeting the stated model rules.
 - ☐ B. It ignores required parts to look tidy.
 - ☐ C. It removes the need for approved spacing in the real project.
 - ☐ D. It proves the exact material and mounting method.
-

4. Revision A was approved, then an important feature changed in B. What is the appropriate response?

- ☐ A. Assume the old approval covers B.
 - ☐ B. Record the change and return the current proposal for review.
 - ☐ C. Delete the revision note.
 - ☐ D. Use whichever file opens first.
-

INVESTIGATE / APPLY

Build a property decision matrix

Fictional materials • practise reasoning, not stock selection

A fictional sign needs to retain its intended flat form, remain manageable within an approved support system and use resources thoughtfully. Compare these imaginary options; they do not describe actual school stock.

Option	Fictional evidence
A	More resistant to bending than B; greater mass; its available layout leaves a connected offcut.
B	Lower mass than A; more flexible; its proposed layout leaves several small separate offcuts.

For each need, identify the relevant property or resource issue and compare the evidence for A and B.

Need	Relevant issue	Evidence comparison
Retain flat form		
Manageability and support		
Resource use		

Recommend one option provisionally. Explain a trade-off and one fact that must be confirmed.

DRAW / EXPLAIN

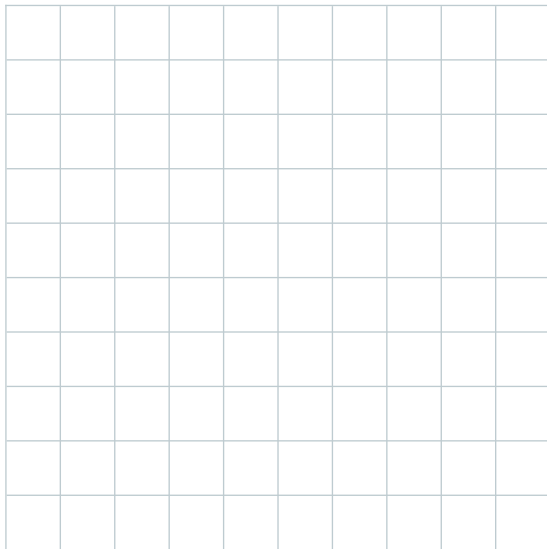
Same coverage, different possibilities

Paper nesting model • no workshop measurements

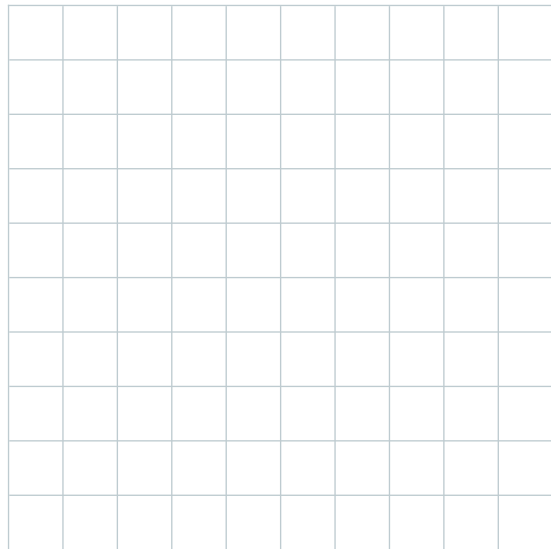
On each ten-by-ten grid, draw six rectangles, each two cells by five cells. Shapes may touch or rotate in this paper model but cannot overlap or cross the boundary. Make two different arrangements. These model rules do not set real project spacing or allowances.

Layouts A and B: mark the six shapes and outline the largest connected unused region in each.

Grid A



Grid B



Calculate the covered and unused percentage for each. Explain why equal coverage does not guarantee equal offcut usefulness.

Recommend one layout using clarity, resource use and one limit. What local information remains unknown?

PROJECT FOLIO

Use the language, explain the decision

Vocabulary puzzle and planning folio

Match each statement to one term: purpose, constraint, vector, trade-off or criterion. Use each term once.

Statement	Term
-----------	------

A design limit that must be respected

Editable image objects described by paths

An agreed statement used to judge success

The job the product is intended to perform

A deliberate compromise between competing benefits

Record the terms in row order, then explain one connection between two of them.

For your own proposal, record the current version, one evidence-based material reason, the teacher review status and one unresolved mounting or layout question.

READ / OBSERVE / APPLY

Sequence the decisions, not just the jobs

Each stage depends on evidence from earlier work.

A production sequence puts stages in a sensible order because some depend on earlier information or results. The design must be reviewed before material is committed. A produced feature needs checking before later work hides or changes its condition. A mounting decision may need attention while the design can still be revised. These dependencies explain the order; a list of jobs alone does not.

The broad course pathway moves from a reviewed proposal to authorised production, quality checks, finishing, approved testing and evaluation. It is a learning sequence, not a machine procedure. The teacher provides the exact local process for the selected project. You should be able to explain why a stage comes first while leaving operating methods and practical settings with the authorised instruction.

Roles make responsibility clear. A student may prepare evidence or record observations; a teacher checks and authorises; an authorised person performs a practical action. The teacher determines the actual allocation. Confidence, experience or watching another person does not create permission. If responsibility is unclear, pause at the hold point and establish who decides and who acts.

A hold point is a planned pause for a proceed, revise or stop decision. Record what was checked, which evidence and version were used, who made the decision and what it covered. Approval does not automatically extend to every later action. If the design or result changes, record the difference and have the current state reviewed. Monitoring compares progress with the approved criteria throughout production. Describe an unexpected result neutrally, identify the affected criterion and refer it to the appropriate person. This makes a timely response possible and creates a truthful account of how quality was controlled.

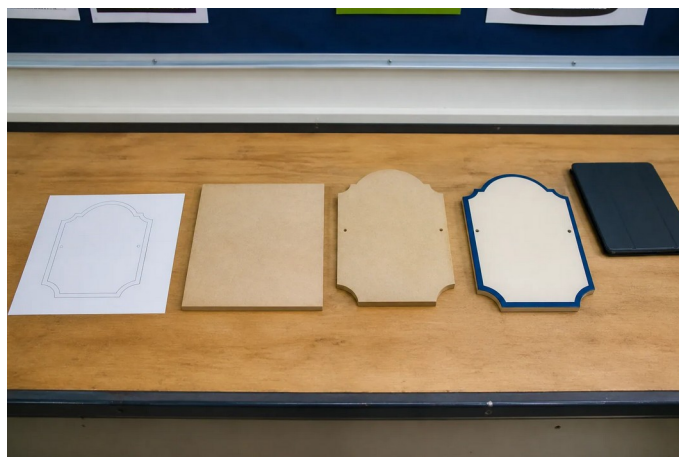


Figure 11. Illustrative production evidence: broad stages connect through checks and decisions. The image is neither an operating procedure nor an actual approval record.

READ / OBSERVE / APPLY

Describe quality with evidence

A precise observation gives the next decision a clear starting point.

Quality means meeting agreed criteria. “Finished” describes progress, while “meets the required features and approved reference” describes a check. A sign may need readable forms, correct orientation, confirmed dimensions and suitable edge and surface condition. The teacher approved brief and plan determine the actual requirements and their priorities. Decorative preference cannot override a safety concern.

A cut or formed edge may show roughness, marks or an unwanted projection called a burr. Recognising such a feature is different from being authorised to inspect it by touch or improve it. Keep hands away from an uncertain edge and refer the observation to the teacher. The teacher supplies the safe inspection method, practical correction, protective equipment and supervision. A photograph does not prove that every edge is safe to handle.

Choose evidence that answers the question being asked. A permitted photograph can show a visible feature and its location. A confirmed measurement compares a size. An approved reference helps establish the target condition. None provides every answer on its own. A useful inspection note names the observed condition, its location, the affected criterion and the decision required. It describes what is known without guessing the cause.

Quality control forms a feedback loop: inspect, decide, act with authority and inspect again. The result may be accepted, corrected through an approved method or referred for a different decision. Rechecking against the same criterion shows whether the response addressed the concern. Finishing is not one universal technique that can be copied between materials. Its purpose is to bring the feature to the required condition under the local process. Recording a correction honestly can show strong understanding, because the evidence explains how the final result was achieved.

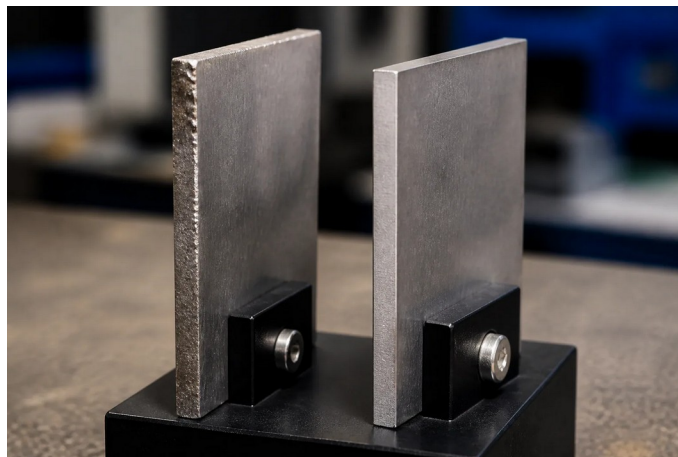


Figure 12. Illustrative edge-quality comparison. A visible projection is evidence to report; the image does not establish safe handling or prescribe a removal method.

READ / OBSERVE / APPLY

Find value in the material record

Compare the planned layout with what actually happened.

Before production, a nested layout predicts how required parts may occupy the approved stock. Afterwards, the material record can show what actually happened. Compare the plan and result using the same categories: required parts, unused area, connected offcuts and any material consumed by rework. A difference is useful evidence. It may reveal an effective layout, an unexpected issue or an opportunity to improve the next proposal.

An offcut is material remaining after parts are produced. It may still have value, but its appearance cannot establish safe handling or reuse. Size, shape, condition, material identity and the school's storage system all matter. Record and refer the information through the teacher's process; do not pick up a piece or place it in a container on your own. A large connected area may be useful for a later approved task, while a small or uncertain piece may need another pathway.

Responsible resource thinking begins with reducing avoidable use through sound planning. It then considers suitable reuse and the approved recycling or disposal system. The aim is useful value while meeting the criteria, not a claim that every gap could have been eliminated. Required spacing, orientation and other process limits still govern what is possible.

Select evidence because it reveals a decision. A dated layout, approval state, actual result, quality note and offcut decision tell more than nearly identical photographs. Caption each item as plan, result or learning model. Keep private names, people and sensitive details out of permitted photographs. To improve, identify a difference, suggest a change, predict its benefit and name a check or trade-off.



Figure 13. Illustrative sorting of offcuts by form. Identity, condition, handling and any reuse decision must be confirmed through the teacher's system.

Fictional paper case: a 100-cell sheet should give six 10-cell shapes and one 40-cell remainder. The result has six correct shapes, a spoiled 10-cell shape, and 20-cell and 10-cell remainders. Compare the useful output and remaining material. Suggest one improvement and a check.

WORKED EXAMPLE

Turn a vague comment into an inspection note

Quality language clinic • fictional observations only

“That edge is bad” is too vague to guide a decision. A stronger fictional note is: “A projection is visible at the lower curve. It differs from the approved edge reference and may affect safe handling. The teacher needs to confirm the inspection and next action.” This names condition, location, criterion and decision without guessing a cause or prescribing a correction.

Vague comment	Fictional information available
“It is crooked.”	The upper feature appears tilted relative to the approved orientation example.
“It is the wrong size.”	A confirmed measurement differs from the value on the reviewed drawing.
“The finish is ugly.”	One area of the outer face has a visible mark; the approved appearance criterion requires an even result.

Rewrite the first comment using all four parts of an inspection note.

Rewrite the second comment without inventing dimensions or tolerances.

Rewrite the third comment, then check that it describes evidence rather than taste.

KNOWLEDGE CHECK

Choose a traceable next action

Circle one answer for each question.

1. Why is a production sequence based on dependencies?

- ☐ A. Every project must use the same equipment.
 - ☐ B. Some stages require information or checked results from earlier stages.
 - ☐ C. The longest task must always come first.
 - ☐ D. Recording decisions removes the need for review.
-

2. An unexpected projection is visible on an edge. What should a student do?

- ☐ A. Touch it to decide whether it is sharp.
 - ☐ B. Copy an online removal method.
 - ☐ C. Keep clear and refer the observation through the teacher's process.
 - ☐ D. Assume it is harmless because it is small.
-

3. Which offcut statement is best supported?

- ☐ A. Every leftover piece is rubbish.
 - ☐ B. Every large piece is automatically safe to reuse.
 - ☐ C. Colour alone proves material identity.
 - ☐ D. Potential value depends on identity, form, condition and the approved local system.
-

4. Which evidence sequence best explains production decisions?

- ☐ A. Current plan, approval state, actual result, quality note and offcut decision
 - ☐ B. Five nearly identical uncaptioned photographs
 - ☐ C. A tool-brand list and unrelated product images
 - ☐ D. Only the final colour preference
-

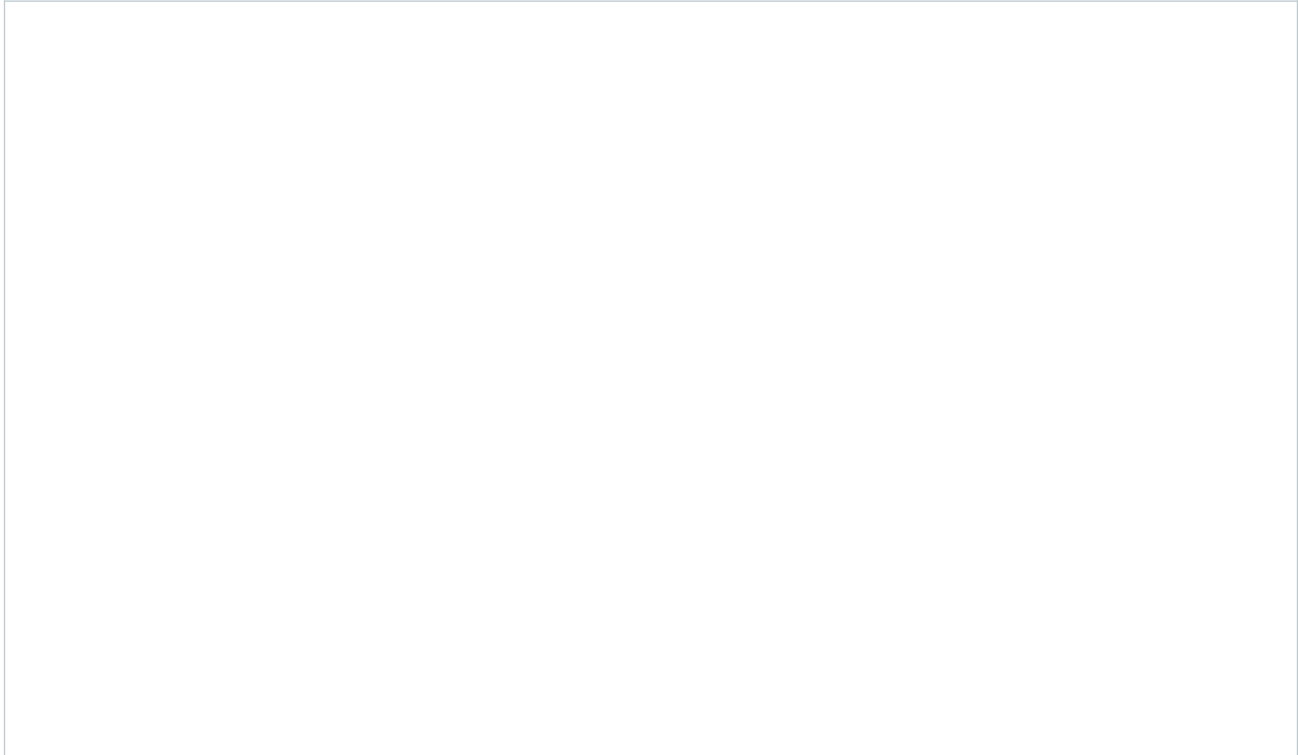
DRAW / EXPLAIN

Build a production-control board

Sequence puzzle • broad learning stages only

Reorder these cards: A — produce through authorised processes and monitor quality; B — test, evaluate and improve; C — define the need and criteria; D — communicate and refine a proposal; E — plan and seek teacher approval.

Arrange the five cards and add dependency arrows. Mark at least three hold points. Label who prepares evidence, who checks or authorises, and who performs practical action.



A design feature changes after review. Show where the board pauses and explain which evidence must return for checking.

INVESTIGATE / APPLY

Edit an evidence story

Fictional records • select five that explain the decisions

Available records: A — dated layout V2; B — tool-brand list; C — recorded teacher decision for V2 and its stated next stage; D — uncaptioned close-up; E — duplicate of D; F — actual arrangement after authorised production using V2; G — favourite-colour list; H — quality note comparing the result with an approved reference; I — image containing unapproved personal information; J — undated old layout; K — offcut record referred to the teacher's system; L — unrelated stock photograph.

Choose five records and write captions that connect plan, approval, result, quality and offcut decision.

Record	Precise caption

Propose one improvement to the evidence system and explain how you would check its benefit.

PROJECT FOLIO

Record the plan, result and response

Folio links • production plan, process and quality

Use your own permitted evidence from one stage. State “planning only” if it has not occurred. Show the relationship between the current plan, what happened, a check and the next authorised decision. Do not photograph people or workshop details without permission.

Sketch or place the evidence. Label its stage, version or sequence, and the feature relevant to the check.

Describe the approved plan and actual result. What criterion was checked, and what did the check show?

Record the decision, who made it and what next action it covered. Add one resource or offcut consideration.

READ / OBSERVE / APPLY

Think of the finish as a system

Compatibility and starting condition influence the result.

A finish is more than its visible colour. The system includes the base material, its starting condition, any approved preparation, coating layers and the environment in which the product will be used. These parts need to work together. A suitable finish may support appearance, protection or cleaning, but its effectiveness cannot be inferred from a colour choice alone.

The substrate is the material or surface receiving a coating. Surface preparation aims to bring it to the condition required by the approved system. Dust, oil, moisture, loose material, corrosion or unevenness may affect spreading, adhesion or appearance. Adhesion means remaining attached to another material. Preparation is not a universal instruction to sand or clean everything: the actual material, residue and product determine the approved treatment and controls.

Different coating layers may have different purposes. A primer is a preparatory coating supporting the approved system; a topcoat provides the intended outer appearance and service surface. A familiar product name does not prove that two layers are compatible with each other or the substrate. The teacher confirms the products and any intermediate stages using the relevant current information. Record those actual sources rather than copying another project's combination.

A coating also needs to develop its intended properties. Dry-to-touch, ready for a later stage and fully cured are not necessarily the same state. Product and conditions affect timing, so use the teacher's instruction and current product information. Judge the eventual result against confirmed criteria such as even appearance, intended coverage and legibility. Sustainability includes useful life, maintenance, avoiding rework and managing products through the school's system. More coating is not automatically better; a reasoned judgement considers the whole compatible system and its intended context.

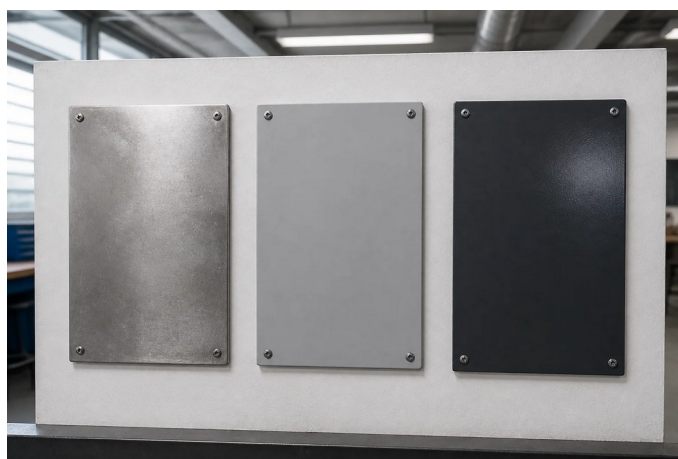


Figure 14. Conceptual surface-system evidence. Substrate, preparation, compatible layers and readiness need confirmation; appearance does not identify an approved product or timing.

READ / OBSERVE / APPLY

Follow the whole support system

Balance and use depend on relationships beyond the visible sign.

A hanging sign works with its connections and support. The sign, connection points, connectors, support and surrounding context form a system. A load path describes the route through which forces travel across those connected parts to the support. A well-made visible sign cannot compensate for an unsuitable connection or support elsewhere in the system.

Gravity acts through the object's centre of mass, whose location depends on how mass is distributed. Connection geometry influences its resting orientation and movement. A symmetrical outline does not guarantee symmetrical mass: openings, attachments or different materials may change the distribution. Stability concerns resistance to unwanted movement or tipping in the intended context. For a suspended sign, relevant movement may include tilt or swing; an alternative freestanding holder raises different questions.

Context changes the criteria. Location, people nearby, weather exposure, visibility, contact and maintenance all matter. A sign intended for one setting cannot automatically be used in another. The teacher confirms the complete mounting and use conditions. For the tea-light alternative, only an approved LED device is within this course. For the pot plant alternative, the intended pot, total approved load, support surface and stability requirements need separate confirmation.

A meaningful test has a confirmed criterion, an approved method, a controlled context and a decision rule. Record what was checked, who authorised or observed it and what the result showed. Students do not invent loads or installation tests. A changed condition, loosened connection or unexpected movement means the test stops and the teacher decides the next action. A successful classroom check supports only the conclusion allowed by its method and context; it does not authorise public installation. Paper models can teach relationships, but their dimensions, support capacity and results cannot be transferred to the actual project.



Figure 15. Illustrative product-and-support concept. The teacher must confirm actual connections, location, loads, test method and permission before hanging, loading or use.

READ / OBSERVE / APPLY

Build a judgement that can be followed

Use criteria, evidence and relevant feedback together.

An evaluation explains how well the product meets its purpose and criteria. A description identifies what is present; an evaluation explains why that matters. “There are two lines of text” is a description. A judgement about their legibility needs evidence from the approved viewing check and a reason connecting the result to the criterion.

Use the strongest evidence available for the question. A teacher-observed controlled test may support a performance judgement more strongly than a preference. A photograph may support a visible observation without proving a hidden connection or long-term durability. Balance strengths and limitations. If a relevant check is absent, say that the criterion is not yet evidenced instead of inventing a positive result.

Classify differences carefully. A defect fails a required criterion. A variation is a difference whose significance still needs judging. A design decision is an intentional choice made for a reason. A changed curve could be any of these depending on its effect and the approved brief. Describe the feature neutrally, identify the criterion and inspect the decision history before choosing a label. Do not call every difference a failure or rename a defect to avoid addressing it.

Feedback is useful when it identifies evidence and a direction for investigation. Consider whether the reviewer understood the user and context, whether the comment connects to a criterion and whether other evidence supports it. Then explain whether you accepted, modified or rejected the suggestion. Prioritise one or two improvements with a clear issue, proposed change, predicted benefit and future check. The change might improve the product, production process or evidence record. Finish with a transferable lesson — a principle that can guide a later task without copying unconfirmed dimensions, materials or procedures.

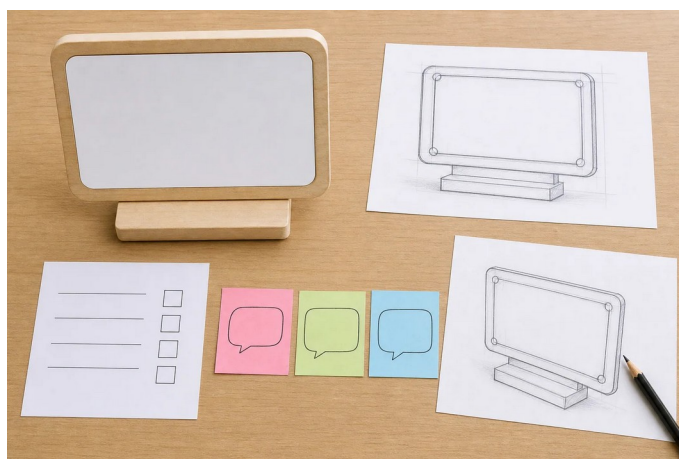


Figure 16. Illustrative evaluation records: a criterion and relevant evidence support a judgement and improvement. No formal assessment decision is represented.

WORKED EXAMPLE

Read the finish evidence carefully

Finish-state puzzle and fictional system audit

Classify each clue using: bare surface, primer stage, topcoat or teacher to confirm. These identify purposes and unknowns, not products for practical use.

Clue	Your classification
Uncoated material before a finish system is selected	
The preparatory layer supporting a compatible system	
The intended outer appearance and service layer	
The actual classroom product and application method	

Record the classifications in row order.

Fictional pack: the intended setting is outdoors. A proposed primer is labelled as preparatory; a proposed topcoat is labelled as the outer layer. Compatibility with the substrate is not recorded. A note says, “It feels dry, so the finish is ready for the approved use test.” No readiness information is supplied.

Identify two supported connections and two unsupported assumptions in the pack.

Write three confirmation questions and one criteria-based inspection statement.

KNOWLEDGE CHECK

Check the system and the conclusion

Circle one answer for each question.

1. Why does “dry-to-touch” not automatically establish readiness for a use test?

- ☐ A. All coatings take the same time.
 - ☐ B. Touching a surface proves its long-term durability.
 - ☐ C. The topcoat chooses the mounting method.
 - ☐ D. Readiness depends on the approved product information, conditions and teacher direction.
-

2. Which statement best explains a load path?

- ☐ A. Forces pass through connected parts towards the support.
 - ☐ B. The file moves from a drawing folder to a printer.
 - ☐ C. The lettering follows a decorative curve.
 - ☐ D. Only the visible sign needs consideration.
-

3. A successful controlled classroom check supports which conclusion?

- ☐ A. The product is approved for every public location.
 - ☐ B. The result met the checked criterion within that approved method and context.
 - ☐ C. Any future load increase is acceptable.
 - ☐ D. All hidden features and future conditions have been verified.
-

4. Which improvement is specific enough to investigate?

- ☐ A. Do everything better.
 - ☐ B. Choose whatever colour a friend prefers.
 - ☐ C. Compare two text spacings earlier against the same approved viewing criterion.
 - ☐ D. Remove records of every revision.
-

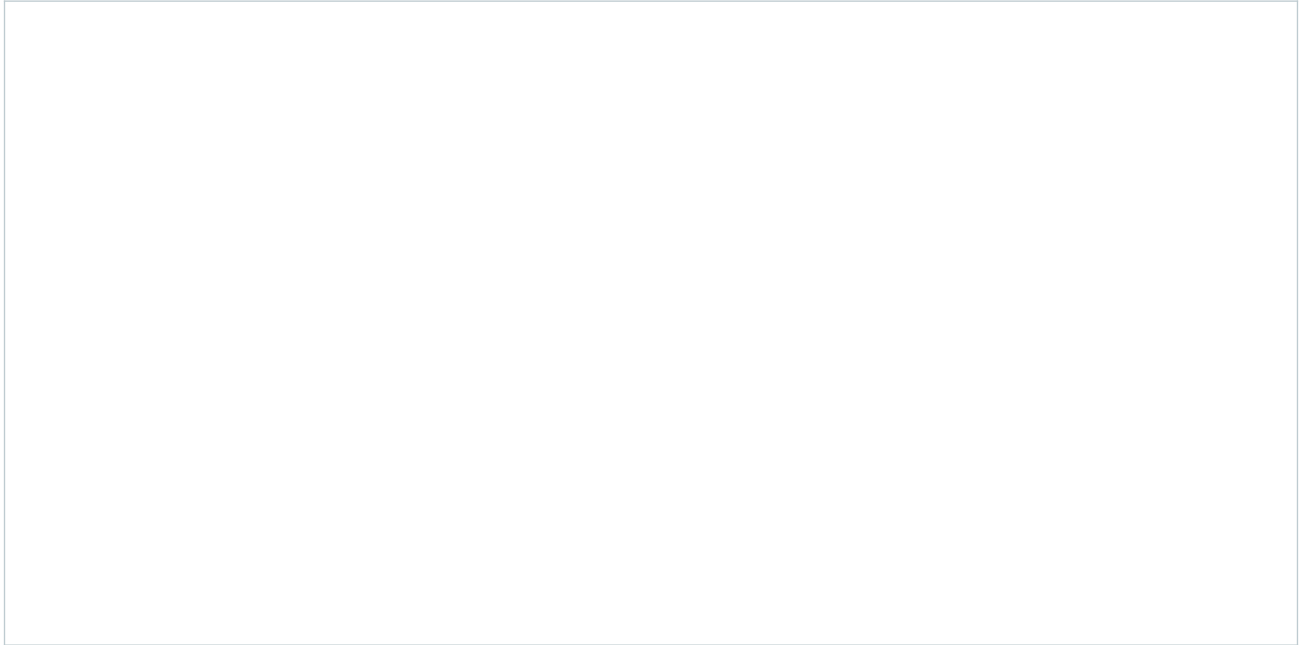
DRAW / EXPLAIN

Reason with a paper balance model

Conceptual forces • no real hanging or loading

Paper case: a straight model bar has a conceptual central suspension point. Two identical cards of equal mass are positioned at equal distances on opposite sides. The supplied result is level. In a revised model, one card has greater mass while positions stay the same. Ignore the bar's mass for this thought experiment.

Draw both stated models. Trace a conceptual load path from each card towards the support and predict the revised resting tendency.



Explain how the change affects balance. Identify one different design change that could also affect orientation.

List facts that cannot be transferred to your actual project, and identify the approval needed before use.

INVESTIGATE / APPLY

Sort claims, then weigh the feedback

Evidence puzzle and an improvement pitch

Classify each statement as evidence, opinion or unknown/confirm. Treat the recorded checks as fictional results within their stated scope.

Statement	Category
The edge matched the approved reference in the recorded check.	
This is the coolest design.	
The support can carry an unapproved load without testing.	
The sign remained level in the approved controlled check.	
Every viewer will prefer this colour.	

Record the categories in order and explain one limit of an evidence statement.

Feedback cards: A — “The lower word was difficult to distinguish in the viewing record; compare spacing.” B — “Use my favourite colour.” C — “The earlier version is missing; preserve it so the change can be checked.” D — “Something is wrong.”

Sort A-D into criterion-linked, preference-only or needing clarification. Identify the strongest two comments.

Write an improvement pitch responding to one strong comment: change, predicted benefit and future check.

PROJECT FOLIO

Make the final judgement traceable

Folio links • finish, function, evaluation and improvement

Use your own evidence from an approved finish or function check. If the check has not happened, label the record “not yet evidenced” and identify what is required. Record the actual criterion, method or instruction reference and teacher review status; do not create a new test.

Show the relevant evidence with its stage and date or sequence label. Annotate the feature the evidence can actually support.

Make a balanced judgement against the confirmed criteria. Include one strength, one limitation and any distinction between a defect, variation or approved decision.

Explain how you used feedback and propose one high-priority improvement. State the benefit, future check and a lesson you could transfer to another project.

PROJECT FOLIO

Set the project direction

Use this record for your actual teacher-confirmed project.

Name your pathway and write the brief: intended audience or user, need, context and proposed response.

Write three criteria. Beside each, state what teacher approved evidence could help judge it.

Criterion	Evidence to seek
1.	
2.	
3.	

Record two confirmed constraints and two details still needing teacher confirmation. Identify the source of each confirmed constraint.

Research link Use your comparison on page 12 and any additional teacher approved research. Record the source and explain how it influenced a decision.

PROJECT FOLIO

Develop two considered options

Work within your teacher-confirmed pathway and record the thinking behind different ideas.

OPTION A — sketch the complete proposal. Label the intended message or use, a form feature and a relevant constraint.

OPTION B — develop a meaningfully different proposal. Change more than colour alone and explain the intended effect.

Compare the options against one criterion. Record useful feedback and the change it suggests.

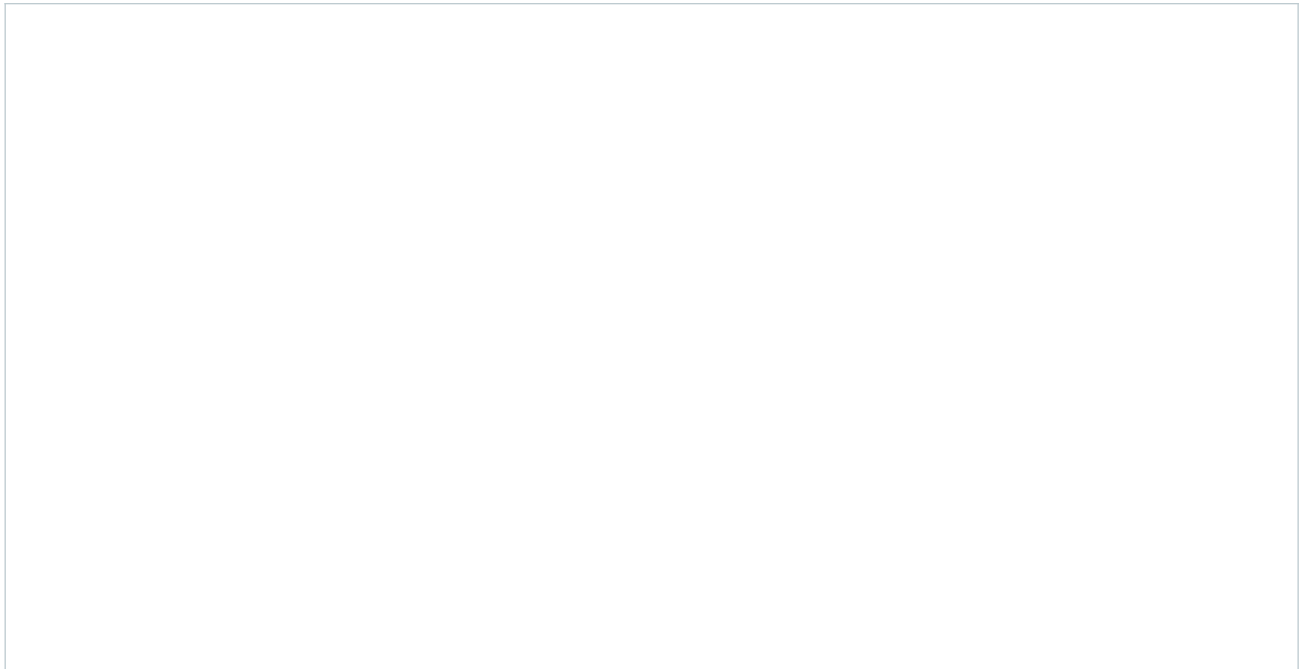
Annotate with purpose Point to a feature, name the choice and explain its intended effect. Leave both large sketch areas for your own ideas.

PROJECT FOLIO

Communicate the selected proposal

Make the design readable and the remaining questions visible.

Sketch or attach a print of the current proposal. Label its purpose, hierarchy or use, material intentions and details awaiting approval.



Record the Onshape document or drawing version and the review feedback it responds to.

Justify one provisional material choice from teacher approved options. Link a relevant property to the intended use and explain a trade-off.

What must the teacher confirm before this proposal can move to production?

PROJECT FOLIO

Plan and record the work

Keep the planned sequence and the actual events distinguishable.

List the broad stages for your approved project. Identify a check or hold point and the evidence needed at each stage.

Broad stage	Check or hold point	Evidence
1.		
2.		
3.		
4.		

Record the date and outcome of teacher review, or write “not yet reviewed”. Note any condition that must be met.

Record one event from authorised work: what happened, the check made, any change and the reason. Include a useful photo reference if available.

Record one resource decision from that event and the evidence behind it.

PROJECT FOLIO

Record quality and function evidence

Use the criterion, the authorised check and the observed result together.

Complete the record after the teacher approved check. Write “not yet checked” where evidence is unavailable.

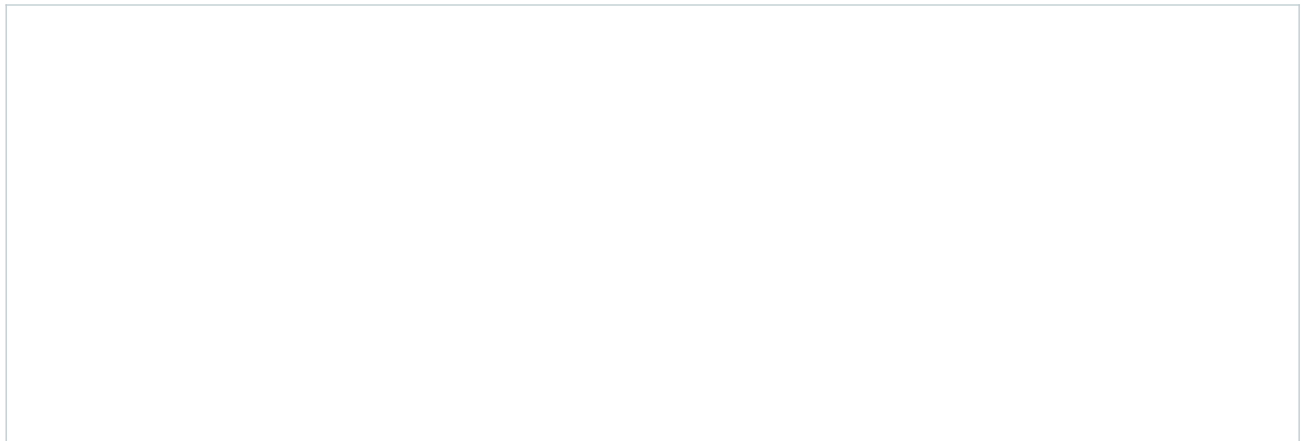
Area	Criterion and approved check	Observed evidence
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Edges and surface

Finish system

Hanging, stability or
intended use

Draw or attach one quality detail, such as teacher-confirmed clearance evidence. Label the location, observation and reference used for comparison.



Explain one quality decision or correction, including who confirmed the next action and what remains uncertain.

PROJECT FOLIO

Evaluate and choose the next improvement

Use specific evidence to make a balanced judgement.

Evaluate your result against two criteria from page 45. Explain one strength and one limitation, using evidence from your checks.

Record one useful feedback point. Explain the improvement you would prioritise, its expected effect and how it could be checked.

What did you learn about making a design decision from evidence?

Sources and next steps

This workbook follows the Swinging Sign guided course and its five modules, learning activities, puzzles and folio. Visit <https://stevencowell.github.io/Swinging-Sign/> for the matching readings and current teacher resources. Use your teacher's current plan and assessment notification for local requirements.

The cover and pathway comparison are original generated illustrations; the technical comparison is an original learning diagram. Other teaching photographs are credited course illustrations. They explain ideas and do not authenticate student work, construction specifications or safe performance.